

Fitting Nonlinear Signal Models Using the Increasing-Data Criterion

Jimei Li  and Feng Ding 

Abstract—To extract important information about the nonlinear signals, this letter makes the utmost of the fitting advantages of Gaussian and polynomial functions, and proposes a nonlinear signal model with broader applications. Then we focus on the parameter estimation issues of the proposed models in the presence of noises. The stability factor recursive algorithm is devised based on the increasing noisy data, which makes full use of the information from the nonlinear signals. Applying the hierarchical identification principle, a two-stage recursive algorithm with higher computational efficiency is developed for the nonlinear signals. The simulation results test the effectiveness of the proposed algorithms from the aspects of estimation accuracy and prediction effect.

Index Terms—Function approximation, nonlinear signal, parameter estimation, recursive search, hierarchical identification.

I. INTRODUCTION

FUNCTION approximation is the core of many engineering research fields such as identification and modeling [1]–[3], which is utilized to reproduce the given input-output relationship of the target object [4]–[6], and to attain a good insight into the behavior of the system under control. Among the common fitting functions, the polynomial function has been confirmed to be a powerful tool to analyze models and fit experimental results [7], [8]. Additionally, the Gaussian function plays an important role in describing various real-world phenomena, peak-like signals and distributions encountered in the experiments [9], [10], but it fails in situations where a signal rises from the baseline and peaks instead of slowly returning [11], [12]. Taking into account this issue, this letter proposes a nonlinear signal model with wider applications, which proves to be a good starting point for nonlinear signal identification.

Estimating the unknown parameters of a model from the observed noisy data, and then using them to describe the underlying signal lies at the heart of many applications in science and engineering [13]–[15]. Some improved approaches and solutions have been proposed to identify the Gaussian and mixed Gaussian functions. The method developed by Al-Nahhal *et*

al. computed the standard deviation directly by exploiting the property of the Gaussian function [16], but it has limitation that it cannot estimate other linear or nonlinear models. Roonizi presented an estimation method for the Gaussian signal riding on a polynomial background [17], which ignores the noises existing in the nonlinear signals, and is susceptible to accumulating the noise error during the numerical integration processes. To avoid the influence of the approximation process on the fitting accuracy, the nonlinear optimization methods were proposed for estimating the noisy Gaussian functions [18].

Differently from the aforementioned methods, this letter utilizes an increasing-data criterion to investigate the parameter estimation issues of nonlinear signals in the presence of noises, which reflects the changes of the nonlinear signals in real time. Moreover, by decomposing the complex optimization issue into a quadratic optimization issue and a nonlinear optimization issue with fewer unknown parameters, this letter proposes a novel hierarchical parameter estimation method with high estimation accuracy and low computational complexity. The proposed methods can combine with other estimation tools [19]–[21], and are extended to estimate linear or nonlinear models with different structures and interfering noises [22], [23].

The letter is organized in the following way. Section II presents a nonlinear signal model with wider applications. Section III proposes a stability factor recursive algorithm for the presented models. A two-stage recursive algorithm is derived in Section IV. Section V performs experimental analyses illustrating the effectiveness of the proposed model and algorithms. Some concluding remarks are given in Section VI.

II. PROBLEM DESCRIPTION

The Gaussian and polynomial functions play an important role in fitting nonlinear signals [24]–[26]. To delve into unknown signals, the nonlinear characteristics need to be extracted from the input-output data. For capturing more features of the nonlinear signals, this letter proposes a model by taking full advantage of the fitting superiorities of Gaussian and polynomial functions, which is given by

$$y_k = \sum_{q=0}^{N-1} \xi_q u_k^q + \zeta e^{-\lambda(u_k - \omega)^2} + v_k, \quad k = 1, 2, \dots, L, \quad (1)$$

where $\{y_k\}$ is a sequence of the observation data disturbed by the white noise v_k , $\{u_k\}$ is a sequence of the given input, $\{v_k\}$ has zero mean and variance σ^2 . Without loss of generality, the initial values are set to be $y_k = 0$ and $v_k = 0$ for $k \leq 0$.

Manuscript received March 24, 2022; revised May 12, 2022; accepted May 12, 2022. Date of publication May 23, 2022; date of current version June 10, 2022. This work was supported by the National Natural Science Foundation of China under Grant 61873111. The associate editor coordinating the review of this manuscript and approving it for publication was Dr. Luis Antonio Azpicueta-Ruiz. (Corresponding author: Feng Ding.)

The authors are with the Key Laboratory of Advanced Process Control for Light Industry (Ministry of Education), School of Internet of Things Engineering, Jiangnan University, Wuxi 214122, China (e-mail: jimeili0625@163.com; fding@jiangnan.edu.cn).

Digital Object Identifier 10.1109/LSP.2022.3177352

The increasing-data criterion is utilized to establish the nonlinear optimization issues in this letter, which utilize more information of the nonlinear signals. Assume that the order N is known, the optimal parameters $\{\xi_q\}_{q=0}^{N-1}$, ζ , ω and λ can be attained by working out the following optimization issue

$$\min_{\{\xi_q\}_{q=0}^{N-1}, \zeta, \omega, \lambda} \sum_{i=1}^k \left\{ y_i - \sum_{q=0}^{N-1} \xi_q u_i^q - \zeta e^{-\lambda(u_i - \omega)^2} \right\}^2. \quad (2)$$

This letter pursues the following strategy in an attempt to solve this issue, i.e., estimating the parameter vector $\vartheta := [\beta^T, \gamma^T]^T \in \mathbb{R}^{N+3}$ based on the observation data y_k , where $\beta := [\xi_0, \xi_1, \dots, \xi_{N-1}, \zeta]^T \in \mathbb{R}^{N+1}$ and $\gamma := [\omega, \lambda]^T \in \mathbb{R}^2$.

III. THE STABILITY FACTOR RECURSIVE ALGORITHM

Let $\hat{\beta}_k := [\hat{\xi}_{0,k}, \hat{\xi}_{1,k}, \dots, \hat{\xi}_{N-1,k}, \hat{\zeta}_k]^T \in \mathbb{R}^{N+1}$, $\hat{\gamma}_k := [\hat{\omega}_k, \hat{\lambda}_k]^T \in \mathbb{R}^2$, and $\hat{\vartheta}_k := [\hat{\beta}_k^T, \hat{\gamma}_k^T]^T \in \mathbb{R}^{N+3}$ denote the estimates of β , γ and ϑ at time k , respectively. Inspired by [27] and applying the nonlinear search to figure out the nonlinear optimization issue in (2), the stability factor recursive (SF-R) algorithm is derived as follows:

$$\hat{\vartheta}_k = \hat{\vartheta}_{k-1} + \frac{[\mathcal{R}_k + \varepsilon \mathbf{I}_{N+3}]^{-1}}{r_{1,k}} \chi(\hat{\vartheta}_{k-1}, k) \eta(\hat{\vartheta}_{k-1}, k), \quad (3)$$

$$r_{1,k} = r_{1,k-1} + \|\chi(\hat{\vartheta}_{k-1}, k)\|^2, \quad r_{1,0} = 1, \quad (4)$$

where the gradient vector is defined as

$$\chi(\hat{\vartheta}_{k-1}, k) := [\psi^T(\hat{\gamma}_{k-1}, k), \hat{\beta}_{k-1}^T \psi'_\omega(\hat{\gamma}_{k-1}, k), \hat{\beta}_{k-1}^T \psi'_\lambda(\hat{\gamma}_{k-1}, k)]^T \in \mathbb{R}^{N+3},$$

$\psi(\hat{\gamma}_{k-1}, k) := [1, u_k, u_k^2, \dots, u_k^{N-1}, e^{-\hat{\lambda}_{k-1}(u_k - \hat{\omega}_{k-1})^2}]^T \in \mathbb{R}^{N+1}$, the stability factor ε satisfies $0 < \varepsilon \leq 1$, $\eta(\hat{\vartheta}_{k-1}, k) \in \mathbb{R}$ denotes the innovation, $\mathbf{I}_{N+3}$ represents an identity matrix of order $N+3$, and $\mathcal{R}_k := [\rho_{ij}(k)] \in \mathbb{R}^{(N+3) \times (N+3)}$ is a matrix consisting of

$$\begin{aligned} \rho_{11}(k) &:= \psi(\hat{\gamma}_{k-1}, k) \psi^T(\hat{\gamma}_{k-1}, k) \in \mathbb{R}^{(N+1) \times (N+1)}, \\ \rho_{12}(k) &= \rho_{21}^T(k) := -\psi'_\omega(\hat{\gamma}_{k-1}, k) \eta(\hat{\vartheta}_{k-1}, k) \\ &\quad + \psi(\hat{\gamma}_{k-1}, k) \hat{\beta}_{k-1}^T \psi'_\omega(\hat{\gamma}_{k-1}, k) \in \mathbb{R}^{N+1}, \\ \rho_{13}(k) &= \rho_{31}^T(k) := -\psi'_\lambda(\hat{\gamma}_{k-1}, k) \eta(\hat{\vartheta}_{k-1}, k) \\ &\quad + \psi(\hat{\gamma}_{k-1}, k) \hat{\beta}_{k-1}^T \psi'_\lambda(\hat{\gamma}_{k-1}, k) \in \mathbb{R}^{N+1}, \\ \rho_{22}(k) &:= -\hat{\beta}_{k-1}^T \psi''_{\omega, \omega}(\hat{\gamma}_{k-1}, k) \eta(\hat{\vartheta}_{k-1}, k) \\ &\quad + \|\psi'_\omega(\hat{\gamma}_{k-1}, k) \circ \hat{\beta}_{k-1}\|^2 \in \mathbb{R}, \\ \rho_{23}(k) &= \rho_{32}(k) := -\hat{\beta}_{k-1}^T \psi''_{\omega, \lambda}(\hat{\gamma}_{k-1}, k) \eta(\hat{\vartheta}_{k-1}, k) \\ &\quad + \hat{\beta}_{k-1}^T \psi'_\omega(\hat{\gamma}_{k-1}, k) [\psi'_\lambda(\hat{\gamma}_{k-1}, k)]^T \hat{\beta}_{k-1} \in \mathbb{R}, \\ \rho_{33}(k) &:= -\hat{\beta}_{k-1}^T \psi''_{\lambda, \lambda}(\hat{\gamma}_{k-1}, k) \eta(\hat{\vartheta}_{k-1}, k) \\ &\quad + \|\psi'_\lambda(\hat{\gamma}_{k-1}, k) \circ \hat{\beta}_{k-1}\|^2 \in \mathbb{R}, \end{aligned}$$

the symbol $\circ$ represents the Hadamard product.

To analyze the convergence of the SF-R algorithm, assume that the noise v_k satisfies $\mathbb{E}[v_k^2] \leq \sigma^2 < \infty$, $\mathbb{E}[v_k] = 0$; the issue in (2) has a unique minimum at $\vartheta = \hat{\vartheta}_k$; $\chi(\hat{\vartheta}_{k-1}, k) \eta(\hat{\vartheta}_{k-1}, k)$ and $\mathcal{R}_k$ exist, and are continuous with respect to ϑ ; $\lim_{k \rightarrow \infty} \mathcal{R}_k \chi(\hat{\vartheta}_{k-1}, k) \eta(\hat{\vartheta}_{k-1}, k)$ exists for any ϑ .

Under the above assumptions, the local extreme value is ϑ , and $\mathcal{R}_k$ is nonsingular. In the neighborhood with radius ϵ centered on the true values, for any initial value $\hat{\vartheta}_0$, the recursive estimate $\hat{\vartheta}_k$ converges to ϑ .

Proof: Firstly, define the estimation error $\mathbf{d}_k := \hat{\vartheta}_k - \vartheta \in \mathbb{R}^{N+3}$ and $\mathbf{g}(\vartheta) := -\chi(\vartheta, k) \eta(\vartheta, k) \in \mathbb{R}^{N+3}$. Expanding $\mathbf{g}(\vartheta)$ at $\vartheta = \hat{\vartheta}_k$ gives

$$\begin{aligned} \mathbf{g}(\vartheta) &= \mathbf{g}(\hat{\vartheta}_k - \mathbf{d}_k) = \mathbf{g}(\hat{\vartheta}_k) - \frac{\partial \mathbf{g}(\hat{\vartheta}_k)}{\partial \vartheta} \mathbf{d}_k + o(\|\mathbf{d}_k\|^2) \\ &= -\chi(\hat{\vartheta}_k, k) \eta(\hat{\vartheta}_k, k) - \mathcal{R}_k \mathbf{d}_k + o(\|\mathbf{d}_k\|^2). \end{aligned}$$

Since ϑ is the extreme value, one can obtain $\mathbf{g}(\vartheta) = \mathbf{0}$, i.e., $-\chi(\hat{\vartheta}_k, k) \eta(\hat{\vartheta}_k, k) - \mathcal{R}_k \mathbf{d}_k + o(\|\mathbf{d}_k\|^2) = \mathbf{0}$. When $\hat{\vartheta}_k$ is sufficiently close to ϑ , and $\mathcal{R}_k$ is nonsingular, pre-multiplying the above equation by $\mathcal{R}_k^{-1}$ gives

$$\begin{aligned} \mathbf{0} &= \mathcal{R}_k^{-1} [-\chi(\hat{\vartheta}_k, k) \eta(\hat{\vartheta}_k, k) - \mathcal{R}_k \mathbf{d}_k + o(\|\mathbf{d}_k\|^2)] \\ &= -[\hat{\vartheta}_{k+1} - \hat{\vartheta}_k] - [\hat{\vartheta}_k - \vartheta] + o(\|\mathbf{d}_k\|^2) \\ &= -\mathbf{d}_{k+1} + o(\|\mathbf{d}_k\|^2). \end{aligned}$$

So there exists a small positive number m such that $\|\mathbf{d}_{k+1}\| \leq m\|\mathbf{d}_k\|^2$, which means that the developed algorithm has the performance of second-order convergence.

Next, we prove $\|\mathbf{d}_{k+1}\| \rightarrow 0$. Here, $\hat{\vartheta}_k$ is sufficiently close to ϑ as $\|\hat{\vartheta}_k - \vartheta\| = \|\mathbf{d}_k\| < \tau$. Then, for any k , we have

$$\|\hat{\vartheta}_{k+1} - \vartheta\| = \|\mathbf{d}_{k+1}\| \leq m\|\mathbf{d}_k\|^2 = m\|\mathbf{d}_k\| \|\mathbf{d}_k\| \leq m\tau \|\mathbf{d}_k\|.$$

Here, one can take τ as a sufficiently small number and make $m\tau < 1$. Hence, $\|\mathbf{d}_{k+1}\| \leq (m\tau)^k \|\mathbf{d}_0\| \rightarrow 0$. The proof is completed.

IV. THE TWO-STAGE RECURSIVE ALGORITHM

By decomposing the complex optimization issue, the two-stage recursive (TS-R) algorithm is developed for the nonlinear signals. The optimization issue for finding the parameter vector β is formulated as $\min_{\beta} \sum_{i=1}^k \{y_i - \psi^T(\gamma, i) \beta\}^2$.

For exploiting the increasing-data criterion and realize online estimation, we apply the recursive technique and derive a linear parameter estimation sub-algorithm for β :

$$\hat{\beta}_k = \hat{\beta}_{k-1} + \mathbf{D}_k [y_k - \psi^T(\gamma, k) \hat{\beta}_{k-1}], \quad (5)$$

$$\mathbf{D}_k = \Sigma_{k-1} \psi(\gamma, k) [1 + \psi^T(\gamma, k) \Sigma_{k-1} \psi(\gamma, k)]^{-1}, \quad (6)$$

$$\Sigma_k = \Sigma_{k-1} - \mathbf{D}_k \psi^T(\gamma, k) \Sigma_{k-1}. \quad (7)$$

Since y_k is highly nonlinear with respect to the parameters ω and λ , we apply the nonlinear search for estimating ω and λ . Let $\hat{\omega}_k$ and $\hat{\lambda}_k$ denote the estimates of ω and λ at time k , respectively.

TABLE I
THE COMPUTATIONAL COST OF THE SF-R AND TS-R ALGORITHMS AT EACH STEP

Algorithms	Expressions	Multiplication	Addition	Total Flop
SF-R	$\hat{\boldsymbol{\theta}}_k = \hat{\boldsymbol{\theta}}_{k-1} + \boldsymbol{\Xi}_k \hat{\boldsymbol{\chi}}_k \eta_k / [r_{1,k-1} + \ \hat{\boldsymbol{\chi}}_k\ ^2] \in \mathbb{R}^{N+3}$	$N^2 + 9N + 18$	$N^2 + 7N + 12$	
	$\boldsymbol{\Xi}_k = [\boldsymbol{\Upsilon}_k + \varepsilon \mathbf{I}_{N+3}]^{-1} \in \mathbb{R}^{(N+3) \times (N+3)}$	$N^3 + 10N^2 + 33N + 36$	$N^2 + 6N + 9$	
	$\boldsymbol{\rho}_{11}(k) = \hat{\boldsymbol{\psi}}_k \hat{\boldsymbol{\psi}}_k^T \in \mathbb{R}^{(N+1) \times (N+1)}$	$N^2 + 2N + 1$	0	
	$\boldsymbol{\rho}_{12}(k) = -\hat{\boldsymbol{\psi}}'_\omega(k) \eta_k + \hat{\boldsymbol{\psi}}_k \hat{\boldsymbol{\beta}}_{k-1}^T \hat{\boldsymbol{\psi}}'_\omega(k) \in \mathbb{R}^{N+1}$	$2N^2 + 5N + 3$	$N^2 + 2N + 1$	
	$\boldsymbol{\rho}_{13}(k) = -\hat{\boldsymbol{\psi}}'_\lambda(k) \eta_k + \hat{\boldsymbol{\psi}}_k \hat{\boldsymbol{\beta}}_{k-1}^T \hat{\boldsymbol{\psi}}'_\lambda(k) \in \mathbb{R}^{N+1}$	$2N^2 + 5N + 3$	$N^2 + 2N + 1$	$N_1 = 2N^3 + 20N^2$
	$\boldsymbol{\rho}_{22}(k) = -\hat{\boldsymbol{\beta}}_{k-1}^T \hat{\boldsymbol{\psi}}''_{\omega,\omega}(k) \eta_k + \ \hat{\boldsymbol{\psi}}'_\omega(k) \circ \hat{\boldsymbol{\beta}}_{k-1}\ ^2 \in \mathbb{R}$	$3N + 4$	$2N + 1$	$+94N + 104$
	$\boldsymbol{\rho}_{23}(k) = -\hat{\boldsymbol{\beta}}_{k-1}^T \hat{\boldsymbol{\psi}}''_{\omega,\lambda}(k) \eta_k + \hat{\boldsymbol{\beta}}_{k-1}^T \hat{\boldsymbol{\psi}}'_\omega(k) [\hat{\boldsymbol{\psi}}'_\lambda(k)]^T \hat{\boldsymbol{\beta}}_{k-1} \in \mathbb{R}$	$4N + 5$	$3N + 1$	
	$\boldsymbol{\rho}_{33}(k) = -\hat{\boldsymbol{\beta}}_{k-1}^T \hat{\boldsymbol{\psi}}''_{\lambda,\lambda}(k) \eta_k + \ \hat{\boldsymbol{\psi}}'_\lambda(k) \circ \hat{\boldsymbol{\beta}}_{k-1}\ ^2 \in \mathbb{R}$	$3N + 4$	$2N + 1$	
	$\hat{\boldsymbol{\chi}}_k = [\hat{\boldsymbol{\psi}}_k, \hat{\boldsymbol{\beta}}_{k-1}^T \hat{\boldsymbol{\psi}}'_\omega(k), \hat{\boldsymbol{\beta}}_{k-1}^T \hat{\boldsymbol{\psi}}'_\lambda(k)]^T \in \mathbb{R}^{N+3}$	$2N + 2$	$2N$	
	$\eta_k = y_k - \hat{\boldsymbol{\psi}}_k^T \hat{\boldsymbol{\beta}}_{k-1} \in \mathbb{R}$	$N + 1$	$N + 1$	
TS-R	$\hat{\boldsymbol{\beta}}_k = \hat{\boldsymbol{\beta}}_{k-1} + \mathbf{D}_k \eta_k \in \mathbb{R}^{N+1}$	$N + 1$	$N + 1$	
	$\mathbf{D}_k = \boldsymbol{\Sigma}_{k-1} \hat{\boldsymbol{\psi}}_k / [1 + \hat{\boldsymbol{\psi}}_k^T \boldsymbol{\Sigma}_{k-1} \hat{\boldsymbol{\psi}}_k] \in \mathbb{R}^{N+1}$	$N^2 + 4N + 3$	$N^2 + 2N + 1$	
	$\boldsymbol{\Sigma}_k = \boldsymbol{\Sigma}_{k-1} - \mathbf{D}_k [\boldsymbol{\Sigma}_{k-1} \hat{\boldsymbol{\psi}}_k]^T \in \mathbb{R}^{(N+1) \times (N+1)}$	$2N^2 + 4N + 2$	$2N^2 + 3N + 1$	
	$\hat{\omega}_k = \hat{\omega}_{k-1} + \hat{\boldsymbol{\beta}}_{k-1}^T \hat{\boldsymbol{\psi}}'_\omega(k) \eta_k / [r_{2,k}(\hat{h}_\omega(k) + \varepsilon)] \in \mathbb{R}$	$N + 4$	$N + 2$	
	$r_{2,k} = r_{2,k-1} + \hat{\boldsymbol{\beta}}_{k-1}^T \hat{\boldsymbol{\psi}}'_\omega(k) \in \mathbb{R}$	$N + 1$	$N + 1$	$N_2 = 6N^2 + 35N + 37$
	$\hat{h}_\omega(k) = -\hat{\boldsymbol{\beta}}_{k-1}^T \hat{\boldsymbol{\psi}}''_{\omega,\omega}(k) \eta_k + \ \hat{\boldsymbol{\psi}}'_\omega(k) \circ \hat{\boldsymbol{\beta}}_{k-1}\ ^2 \in \mathbb{R}$	$3N + 4$	$2N + 1$	
	$\hat{\lambda}_k = \hat{\lambda}_{k-1} + \hat{\boldsymbol{\beta}}_{k-1}^T \hat{\boldsymbol{\psi}}'_\lambda(k) \eta_k / [r_{3,k}(\hat{h}_\lambda(k) + \varepsilon)] \in \mathbb{R}$	$N + 4$	$N + 2$	
	$r_{3,k} = r_{3,k-1} + \hat{\boldsymbol{\beta}}_{k-1}^T \hat{\boldsymbol{\psi}}'_\lambda(k) \in \mathbb{R}$	$N + 1$	$N + 1$	
	$\hat{h}_\lambda(k) = -\hat{\boldsymbol{\beta}}_{k-1}^T \hat{\boldsymbol{\psi}}''_{\lambda,\lambda}(k) \eta_k + \ \hat{\boldsymbol{\psi}}'_\lambda(k) \circ \hat{\boldsymbol{\beta}}_{k-1}\ ^2 \in \mathbb{R}$	$3N + 4$	$2N + 1$	
	$\eta_k = y_k - \hat{\boldsymbol{\psi}}_k^T \hat{\boldsymbol{\beta}}_{k-1} \in \mathbb{R}$	$N + 1$	$N + 1$	

Applying the nonlinear search, we derive a SF-R sub-algorithm for estimating ω and λ :

$$\hat{\omega}_k = \hat{\omega}_{k-1} + \frac{\boldsymbol{\beta}^T \hat{\boldsymbol{\psi}}'_\omega(\hat{\omega}_{k-1}, \lambda, k) \eta(\boldsymbol{\beta}, \hat{\omega}_{k-1}, \lambda, k)}{r_{2,k}[\hat{h}_\omega(\boldsymbol{\beta}, \hat{\omega}_{k-1}, \lambda, k) + \varepsilon]}, \quad (8)$$

$$r_{2,k} = r_{2,k-1} + [\boldsymbol{\beta}^T \hat{\boldsymbol{\psi}}'_\omega(\hat{\omega}_{k-1}, \lambda, k)]^2, \quad r_{2,0} = 1, \quad (9)$$

$$\hat{\lambda}_k = \hat{\lambda}_{k-1} + \frac{\boldsymbol{\beta}^T \hat{\boldsymbol{\psi}}'_\lambda(\omega, \hat{\lambda}_{k-1}, k) \eta(\boldsymbol{\beta}, \omega, \hat{\lambda}_{k-1}, k)}{r_{3,k}[\hat{h}_\lambda(\boldsymbol{\beta}, \omega, \hat{\lambda}_{k-1}, k) + \varepsilon]}, \quad (10)$$

$$r_{3,k} = r_{3,k-1} + [\boldsymbol{\beta}^T \hat{\boldsymbol{\psi}}'_\lambda(\omega, \hat{\lambda}_{k-1}, k)]^2, \quad r_{3,0} = 1, \quad (11)$$

where

$$\begin{aligned} h_\omega(\boldsymbol{\beta}, \hat{\omega}_{k-1}, \lambda, k) &:= -\boldsymbol{\beta}^T \hat{\boldsymbol{\psi}}''_{\omega,\omega}(\hat{\omega}_{k-1}, \lambda, k) \eta(\boldsymbol{\beta}, \hat{\omega}_{k-1}, \lambda, k) \\ &\quad + \|\hat{\boldsymbol{\psi}}'_\omega(\hat{\omega}_{k-1}, \lambda, k) \circ \boldsymbol{\beta}\|^2 \in \mathbb{R}, \\ h_\lambda(\boldsymbol{\beta}, \omega, \hat{\lambda}_{k-1}, k) &:= -\boldsymbol{\beta}^T \hat{\boldsymbol{\psi}}''_{\lambda,\lambda}(\omega, \hat{\lambda}_{k-1}, k) \eta(\boldsymbol{\beta}, \omega, \hat{\lambda}_{k-1}, k) \\ &\quad + \|\hat{\boldsymbol{\psi}}'_\lambda(\omega, \hat{\lambda}_{k-1}, k) \circ \boldsymbol{\beta}\|^2 \in \mathbb{R}. \end{aligned}$$

It should be noticed that the sub-algorithm in (5)–(7) and the SF-R sub-algorithm in (8)–(11) cannot realize due to the unknown parameters. Inspired by the interactive estimation theory [28]–[30], we replace $\boldsymbol{\beta}$, ω and λ with their corresponding estimates $\hat{\boldsymbol{\beta}}_{k-1}$, $\hat{\omega}_{k-1}$ and $\hat{\lambda}_{k-1}$ at the preceding time to develop the TS-R algorithm for the nonlinear signals.

The computational efficiency of an algorithm can be evaluated by the flops, and an addition or a multiplication counts as a flop. Table I lists the number of flops of the SF-R and TS-R algorithms at each step for computational comparison. The difference between the computation costs of these algorithms at each recursion is $\Delta N := N_1 - N_2 = 2N^3 + 14N^2 + 59N + 65 > 0$. It is clear that the TS-R algorithm can substantially reduce the computational complexity compared with the SF-R algorithm.

For the nonlinear signals with higher orders, the advantage of the TS-R algorithm is even more prominent.

V. EXAMPLES

A. Example 1

Consider: $y_k = \sum_{q=0}^2 \xi_q u_k^q + \zeta e^{-\lambda(u_k - \omega)^2} + v_k$ as the nonlinear signal model. For the comparison purpose, the following algorithms are utilized to estimate the above nonlinear model based on the observed data: i) the FF-MISG algorithm in [18]; ii) the SF-R algorithm; iii) the TS-R algorithm. The parameters are given by $\xi_0 = 0.50$, $\xi_1 = 0.10$, $\xi_2 = 0.22$, $\zeta = 1.20$, $\omega = 0.95$ and $\lambda = 0.32$. The input $\{u_k\}$ is taken as a uniformly distributed random signal sequence between the intervals $[-2, 2]$ with zero mean, the disturbance $\{v_k\}$ is taken as a white noise sequence with zero mean and variance σ^2 . The estimation error $\delta := \|\hat{\boldsymbol{\theta}}_k - \boldsymbol{\theta}\| / \|\boldsymbol{\theta}\|$ with $\boldsymbol{\theta} := [\xi_0, \xi_1, \xi_2, \zeta, \omega, \lambda]^T$ is used to measure the performance of the proposed algorithms against the noise perturbation. Set $L = 3000$ as the data length for model validation.

This letter takes $\varepsilon = 0.10$ to make the SF-R and TS-R algorithms stable. For these algorithms, an appropriate ε can get high parameter estimation accuracy. The value of ε depends on practical experience and requires many tests.

Fixing $\sigma^2 = 0.10^2$, the forgetting factor and the innovation length are taken as 0.80 and 3, respectively. The FF-MISG, SF-R and TS-R estimation errors versus k are shown in Fig. 1, it can be seen that: the parameter estimation errors decrease with k increasing; the proposed SF-R and TS-R algorithms obtain more accurate parameter estimates than the FF-MISG algorithm; among them, the TS-R algorithm has the highest parameter estimation accuracy.

Additionally, the comparison between the actual outputs and the estimated outputs of the SF-R and TS-R algorithms under

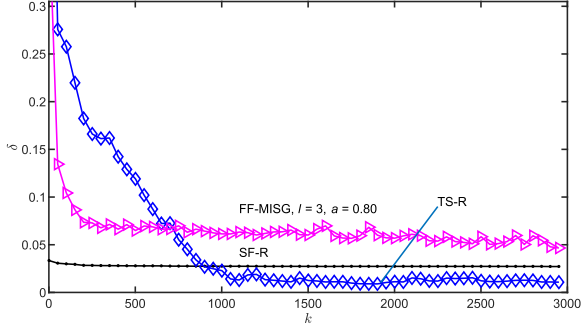
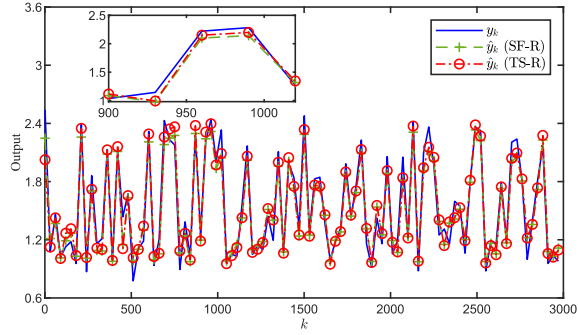
Fig. 1. The FF-MISG, SF-R and TS-R estimation errors against k .

Fig. 2. The contrast of actual outputs and estimated outputs.

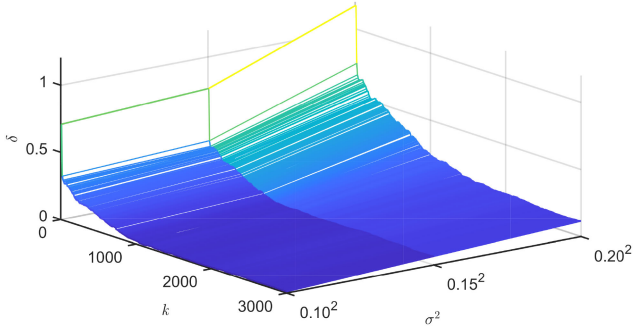


Fig. 3. The TS-R estimation errors with different noise variances.

$\sigma^2 = 0.10^2$ is depicted in Fig. 2. To study the influence of different noises on parameter estimation, we apply the TS-R algorithm with different σ^2 to estimate the nonlinear signal. The results are exhibited in Fig. 3. The above results indicate that the estimated outputs are very close to the actual outputs during the estimation process; and the TR-S algorithm can obtain satisfactory identification results under low noise-to-signal ratio.

B. Example 2

Raman spectra has been used extensively within the realm of biological applications, which is often accompanied by a broad background signal [31]. The nonlinear signal model based on the FF-MISG, SF-R and TS-R algorithms is used to fit the Raman spectra (variation in the intensity of D band and G band) of FePc/defective graphene, the data come from the work [32]. To facilitate subsequent data processing, the normalized data are obtained by using the zscore function in Matlab. Select 3000

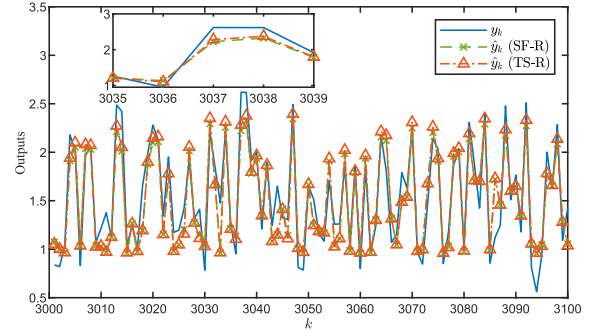


Fig. 4. The contrast of actual outputs and predicted outputs.

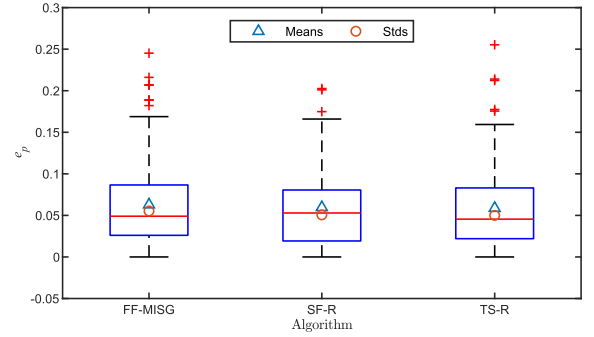


Fig. 5. The distribution of predicted output errors with different algorithms.

data from the normalized data to estimate the nonlinear signal model and 100 data for test and prediction.

The actual outputs and the predicted outputs of the SF-R and TS-R algorithms are depicted in Fig. 4. Additionally, the predicted output error $e_p := \|y_k - \hat{y}_k\|/\|y_k\|$ is defined to demonstrate the forecast results. The distribution of e_p under different algorithms is plotted in Fig. 5. The above results reveal that the SF-R and TS-R algorithms attain good prediction effect, and the nonlinear signal model can fit the Raman spectra very well.

VI. CONCLUSION

This letter proposes a nonlinear signal model with wider applications by utilizing the fitting superiorities of Gaussian and polynomial functions. Then a SF-R algorithm is presented via the increasing noisy data, which can utilize more information of the nonlinear signals. Applying the hierarchical identification principle, this letter derives a TS-R algorithm with high computational efficiency for the nonlinear signals. The simulation results demonstrate the effectiveness of the proposed model and algorithms. The future work will focus on the nonlinear signal modeling and estimation issues.

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